

E - Unbalanced ABC Substrings

分值: 450 分

Problem Statement

You are given a string S of length N consisting of the three characters A, B, and C.

There are $\frac{N(N+1)}{2}$ non-empty substrings of S ; find how many of them satisfy the following condition:

给定一个长度为 N 的字符串 S ，其仅包含三种字符 A、B 和 C。字符串 S 共有 $\frac{N(N+1)}{2}$ 个非空子串，请求出其中满足以下条件的子串数量：

- The numbers of occurrences of A, B, and C are all distinct from each other.
- A、B 和 C 三者的出现次数互不相同。

Count two substrings separately if they are taken from different positions in S , even if they are identical as strings.

即使两个子串的内容完全相同，只要它们在 S 中的位置不同，就需要被单独计数。

What is a substring?

什么是子串？

A **substring** of S is a string obtained by deleting zero or more characters from the beginning and zero or more characters from the end of S . For example, AB is a substring of ABC, but AC is not a substring of ABC.

字符串 S 的**子串**是指：通过删除 S 开头的零个或多个字符，以及结尾的零个或多个字符后得到的字符串。例如，AB 是 ABC 的子串，但 AC 不是 ABC 的子串。

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $|S| = N$
- N is an integer.

- S is a string consisting of the three characters A, B, and C.
- $1 \leq N \leq 2 \times 10^5$
- $|S| = N$
- N 是一个整数。
- S 是一个仅由 A、B 和 C 三种字符组成的字符串。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N

S

Output

Output the answer on a single line.

将答案输出在一行中。

Sample Input 1

6

AABBCC

Sample Output 1

4

Deleting 0 characters from the beginning and 3 characters from the end of S gives AAB, which satisfies the condition.

Deleting 1 character from the beginning and 2 characters from the end of S gives ABB, which satisfies the condition.

Deleting 2 characters from the beginning and 1 character from the end of S gives BBC, which satisfies the condition.

Deleting 3 characters from the beginning and 0 characters from the end of S gives BCC, which satisfies the condition.

No other substrings satisfy the condition.

删除 S 开头的 0 个字符和结尾的 3 个字符，得到的 AAB 满足条件。删除 S 开头的 1 个字符和结尾的 2 个字符，得到的 ABB 满足条件。删除 S 开头的 2 个字符和结尾的 1 个字符，得到的 BBC 满足条件。删除 S 开头的 3 个字符和结尾的 0 个字符，得到的 BCC 满足条件。没有其他子串满足条件。

Sample Input 2

6

ABCABC

Sample Output 2

0

No substring satisfies the condition.

没有子串满足条件。

Sample Input 3

10

ACABCAABAB

Sample Output 3

17